Course Content

In this course we will cover the foundational mathematical ideas necessary for the study of calculus. We will also work on developing a productive mathematical disposition, including persistence in problem-solving, evaluating the reasoning of others, and formulating mathematical justifications. Targeted toward students continuing to MTH 201, this course does not fulfill the core requirement.

By the end of the semester you will be able to...

- Analyze functions defined graphically, numerically, or by formula, including exponential, logarithmic, trigonometric, polynomial, and rational families of functions.
- Perform algebraic computations efficiently and correctly; particularly on exponential, logarithmic, and fractional expressions.
- Construct graphs of trigonometric functions using the properties of the unit circle.
- Use trigonometric, rational, exponential, and logarithmic equations to model situations and solve problems.

Learning Targets

The bulk of the course content is represented by the following specific topics which we call Learning Targets. Your goal over the course of the semester is to develop your understanding of these topics and to demonstrate that understanding via various assessment tasks – primarily our weekly check-ins. These topics are generally aligned with the indicated sections of our book.

Algebra

A1 Solving equations and inequalities I can use algebraic operations to solve equations and evaluate inequalities.

A2 Factoring, multiplying, and rational expressions I can rewrite rational and polynomial expressions through factoring, polynomial multiplication, and simplifying fractional expressions.

Functions

F1 Inputs and outputs I can use function notation, find and simplify the output of a function given an input, and find an input that produces a given output; I can do this for functions given by a formula, a table, and a graph. (1.1 & 1.2)

F2 Rate of change I can find the average rate of change in a function on a given interval and interpret the meaning. (1.3)

F3 Linear and quadratic functions I can model linear and quadratic relationships with equations and find key features graphically and algebraically. (1.4 & 1.5)

F4 Compositions I can find the composition of functions given graphs, tables, and contexts, and interpret the meaning (1.6)

F5 Inverses I can determine whether a function has an inverse and find the formula of the inverse of basic functions that have an inverse. (1.7)

F6 Translations and piecewise functions I can translate graphs functions and construct piecewise functions. (1.8 & 1.9)

F7 Function review I can interpret, represent, and evaluate problems using all previous function learning objectives.

Trigonoetry

T1 Unit circle and arc length I can generate graphs of a circular functions given appropriate information about the circle being traversed, and describe key features of the function. (2.1, 2.2, & 2.3)

T2 Sinusoidal functions I can find the amplitude, period, and midline of a transformed version of the basic sine or cosine function; and I can find the formulas for transformed versions of the basic sine or cosine function that have certain described properties. (2.4)

T3 Right Triangles Given a right triangle with partial information about its sides and angles, I can determine the missing information for the remaining sides and angles, even if the quantities involve an unknown variable. (4.1)

T4 Finding Angles I can interpret inverse trigonometric functions and use them to find angles. (4.3 & 4.4)

T5. Other Trig Functions I can define tangent and reciprocal trigonometric functions and their graphs. (4.2 & 4.5)

T6 Trig Review I can interpret, represent, and evaluate problems using all previous exponential and log learning objectives.

Exponentials and Logarithms

E1 Exponent Properties I can simplify exponential expressions and solve equations using properties of exponents. (3.0)

E2 Exponential Functions I can construct graphs and equations of exponential and logarithmic functions given contextual data or two point values. (3.2 & 3.2)

E3 Equations with Exponents and Logs I can simplify logarithmic expressions using properties of logarithms and find exact solutions to equations involving exponential and logarithmic expressions. (3.4, 3.5, 3.6)

E4 Exponential Review I can interpret, represent, and evaluate problems using all previous exponential and log learning objectives.

Rational Functions and Polynomials

R1 Polynomials and Power Functions I can identify the degree, zeros, turning points, and long-range behavior of a polynomial given its formula or graph, and I can find the formula of a polynomial that fits given data about its behavior. (5.1, 5.2, & 5.3)

R2 Rational Functions I can identify important features of rational functions, such as the asymptotes, end behavior, intercepts and holes. I can find the formula of a rational function that fits given data about its behavior. (5.4 & 5.5)

Mathematical Practice

While not directly tied to specific topics, we will also work on developing more general aspects of a mathematical practice including:

- Ability to communicate mathematical thinking effectively, including correct use of vocabulary, notation, and mathematical representations.
- Ability to develop strategies for solving complex or unfamilar problems.
- Ability to use technology effectively.